Here is the complete, detailed Configuration and Troubleshooting Document for your Splunk Distributed SIEM Lab. You can save this as a Markdown file or copy it into Word/Confluence for your project documentation.

Splunk Distributed SIEM Lab - Configuration & Build Document

Environment: EVE-NG (Virtual Lab)

Architecture: Distributed Splunk Enterprise (Cluster Manager, Indexers, Search Head, Deployment Server, Rsyslog Relay)

Objective: Build a functional SIEM capable of ingesting Linux authentication logs, detecting SSH brute-force attacks, and providing a foundation for SOAR integration.

1. Infrastructure & IP Addressing

All VMs are connected to a central switch in EVE-NG.

Hostname	Role	IP Address	OS	Specs
`Splunk-Master`	Cluster Manager	`192.168.1.10`	Ubuntu 20.04	2 vCPU, 4GB RAM
`Splunk-Idx-01`	Indexer Peer	`192.168.1.11`	Ubuntu 20.04	2 vCPU, 4GB RAM
`Splunk-Idx-02`	Indexer Peer	`192.168.1.12`	Ubuntu 20.04	2 vCPU, 4GB RAM
`Splunk-SH-01`	Search Head	`192.168.1.20`	Ubuntu 20.04	2 vCPU, 4GB RAM
`Splunk-DS-01`	Deployment Server	`192.168.1.30`	Ubuntu 20.04	1 vCPU, 2GB RAM
`splunk-rsyslog01`	Rsyslog Relay & UF	`192.168.1.35`	Ubuntu 20.04	1 vCPU, 2GB RAM
`Ubuntu-Target`	Log Source & UF	`192.168.1.50`	Ubuntu 20.04	1 vCPU, 2GB RAM
`Kali-Linux`	Attacker	`192.168.1.142`	Kali Linux	1 vCPU, 2GB RAM
`Windows-Target`	Log Source	`DHCP / Static`	Windows 10	2 vCPU, 4GB RAM

Hostname & DNS Configuration (All VMs)

To ensure clean communication, hostnames must be updated and mapped in `/etc/hosts`.

1. Set hostname: `sudo hostnamectl set-hostname <hostname>`
2. Edit `/etc/hosts`: `sudo nano /etc/hosts` and add all lab VMs.
3. **Crucial:** Prevent Cloud-Init from reverting the hostname: `sudo sed -i 's/preserve_hostname: false/preserve_hostname: true/' /etc/cloud/cloud.cfg`

2. Splunk Enterprise Installation (Master, Idx-01, Idx-02, SH-01, DS-01)

Run on all 5 Splunk Enterprise VMs:

```
wget -O splunk.deb 'https://download.splunk.com/products/splunk/releases/10.0.2/linux/splunk-10.0.2-e2d18b4767e9-linux-2.6-amd64.deb'
sudo dpkg -i splunk.deb
sudo /opt/splunk/bin/splunk start --accept-license
# Set admin username: admin, password: Changeme123
sudo /opt/splunk/bin/splunk enable boot-start -user splunk
```

Cluster Configuration Sequence

A. On `Splunk-Master` (Cluster Manager):

```
sudo /opt/splunk/bin/splunk edit cluster-config -mode manager -secret isClusterSecret -auth admin:Changeme123
sudo /opt/splunk/bin/splunk restart
```

B. On `Splunk-Idx-01` & `Splunk-Idx-02` (Indexers):

```
sudo /opt/splunk/bin/splunk edit cluster-config -mode peer -manager_uri https://192.168.1.10:8089 -replication_port 9887 -secret isClusterSecret -auth admin:Changeme123
sudo /opt/splunk/bin/splunk restart
```

C. On `Splunk-SH-01` (Search Head):

```
sudo /opt/splunk/bin/splunk edit cluster-config -mode searchhead -manager_uri https://192.168.1.10:8089 -secret isClusterSecret -auth admin:Changeme123
sudo /opt/splunk/bin/splunk restart
```

D. Verify Cluster Health on Master:

```
sudo /opt/splunk/bin/splunk show cluster-status -auth admin:Changeme123
# MUST SHOW: "Indexing Ready YES", "Replication factor met", "Search factor met"
```

---

3. Rsyslog Relay Configuration (`splunk-rsyslog01`)

The relay receives syslog from network devices and log sources, saves them to disk, and forwards them to Splunk.

4. **Edit `/etc/rsyslog.conf`:**

Uncomment `module(load="imudp")`, `input(type="imudp" port="514")`, `module(load="imtcp")`, `input(type="imtcp" port="514")`.

5. **Create `/etc/rsyslog.d/remote.conf`:**

```
$template RemoteLogs,"/var/log/remote/%HOSTNAME%.log"
*.* ?RemoteLogs
```

6. **Set Permissions & Firewall:**

```
sudo mkdir -p /var/log/remote
sudo chown syslog:adm /var/log/remote
sudo chmod 755 /var/log/remote
sudo ufw allow 514/udp
sudo ufw allow 514/tcp
sudo systemctl restart rsyslog
```

7. **Install Splunk Universal Forwarder (UF):**

```
wget -O splunkforwarder.deb "https://download.splunk.com/products/universalforwarder/releases/9.4.0/linux/splunkforwarder-9.4.0-6b4ebe426ca6-linux-amd64.deb"
sudo dpkg -i splunkforwarder.deb
```

```
sudo /opt/splunkforwarder/bin/splunk start --accept-license
sudo /opt/splunkforwarder/bin/splunk enable boot-start
```

8. **Create `inputs.conf` (`/opt/splunkforwarder/etc/system/local/inputs.conf`):**

```
[monitor:///var/log/remote/]
disabled = false
sourcetype = syslog
index = main
```

9. **Create `outputs.conf` (`/opt/splunkforwarder/etc/system/local/outputs.conf`):**

```
[tcpout]
defaultGroup = indexer_cluster

[tcpout:indexer_cluster]
server = 192.168.1.11:9997, 192.168.1.12:9997
```

Run `sudo /opt/splunkforwarder/bin/splunk restart`

---

4. Log Source Configuration (`Ubuntu-Target`)

This VM simulates a target server generating authentication logs.

10. **Install Universal Forwarder:** (Same commands as Step 3.4).
11. **Configure `inputs.conf`:**

```
[monitor:///var/log/auth.log]
disabled = false
sourcetype = linux_secure
index = main
```

12. **Configure `outputs.conf` (Point to Relay):**

```
[tcpout]
defaultGroup = relay_server

[tcpout:relay_server]
server = 192.168.1.35:9997
```

13. **Configure Rsyslog on Ubuntu Target to send logs to the Relay:**

```
echo "*.* @@192.168.1.35:514" | sudo tee /etc/rsyslog.d/50-default.conf
sudo systemctl restart rsyslog
sudo /opt/splunkforwarder/bin/splunk restart
```

14. **Ensure SSH is running on Ubuntu (for the attack):**

```
sudo apt install openssh-server -y
sudo systemctl start ssh
```

---

5. Attack Simulation (`Kali-Linux`)

Generate security events to test the SIEM.

15. **Unzip Wordlist:**

```
sudo gunzip /usr/share/wordlists/rockyou.txt.gz
```

16. **Run Hydra Brute Force:**

```
hydra -l root -P /usr/share/wordlists/rockyou.txt -t 4 ssh://192.168.1.50
```

(Press `Ctrl+C` after 30 seconds. The goal is to generate logs, not crack the password).

🔍 6. Validation & Splunk Search

Go to `http://192.168.1.20:8000` (Search Head).

Search for the Attack:

```
index=* host=ubuntu-target "Failed password"
```

Create the Detection Alert:

```
index=* host=ubuntu-target "Failed password"
| rex field=_raw "from (?<src_ip>\d+\.\d+\.\d+\.\d+)"
| stats count by src_ip
| where count > 5
```

(Save As -> Alert -> Scheduled -> Trigger when results > 0)

Screenshots Splunk VM's Configurations with results..

1- Splunk Master

root@splunk-master: ~

```
root@splunk-master:~#
root@splunk-master:~#
root@splunk-master:~#
root@splunk-master:~# hostname
splunk-master
root@splunk-master:~# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP             192.168.1.10/24 2001:8f8:1a69:3021:b882:cac5:7b60:f8fe/128 f
root@splunk-master:~#
root@splunk-master:~#
root@splunk-master:~# ip route
default via 192.168.1.1 dev eth0 proto static
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.10
root@splunk-master:~# cat /etc/resolv.conf
# This file is managed by man:systemd-resolved(8). Do not edit.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs must not access this file directly, but only through the
# symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a different way,
# replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0
root@splunk-master:~#
root@splunk-master:~# ping -c 2 192.168.1.50 #ping the target ubuntu
PING 192.168.1.50 (192.168.1.50) 56(84) bytes of data.
64 bytes from 192.168.1.50: icmp_seq=1 ttl=64 time=4.13 ms
64 bytes from 192.168.1.50: icmp_seq=2 ttl=64 time=1.20 ms

--- 192.168.1.50 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/mdev = 1.196/2.661/4.126/1.465 ms
root@splunk-master:~#
root@splunk-master:~#
root@splunk-master:~# sudo /opt/splunk/bin/splunk show cluster-status -auth admin:[redacted]
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslConf
etails.

 Replication factor met
 Search factor met
 All data is searchable
 Indexing Ready YES
 HA Mode: Disabled

 splunk-idx-01   4EBB7C97-57F3-48EA-B32F-7D9D2F802F70    default
         Searchable YES
         Status  Up
         Bucket Count=31

 splunk-idx-02   F2EDB6BC-74AB-4C49-8710-CBC93172D4DA    default
         Searchable YES
```

Splunk Rsyslog

```
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~# hostname
splunk-rsyslog01
root@splunk-rsyslog01:~# ip -br a
lo               UNKNOWN        127.0.0.1/8 ::1/128
eth0             UP             192.168.1.35/24 fe80::250:ff:fe00:a00/64
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~# ip route
default via 192.168.1.1 dev eth0 proto static
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.35
root@splunk-rsyslog01:~# cat /etc/resolv.conf
# This file is managed by man:systemd-resolved(8). Do not edit.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs must not access this file directly, but only through the
# symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a different way,
# replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

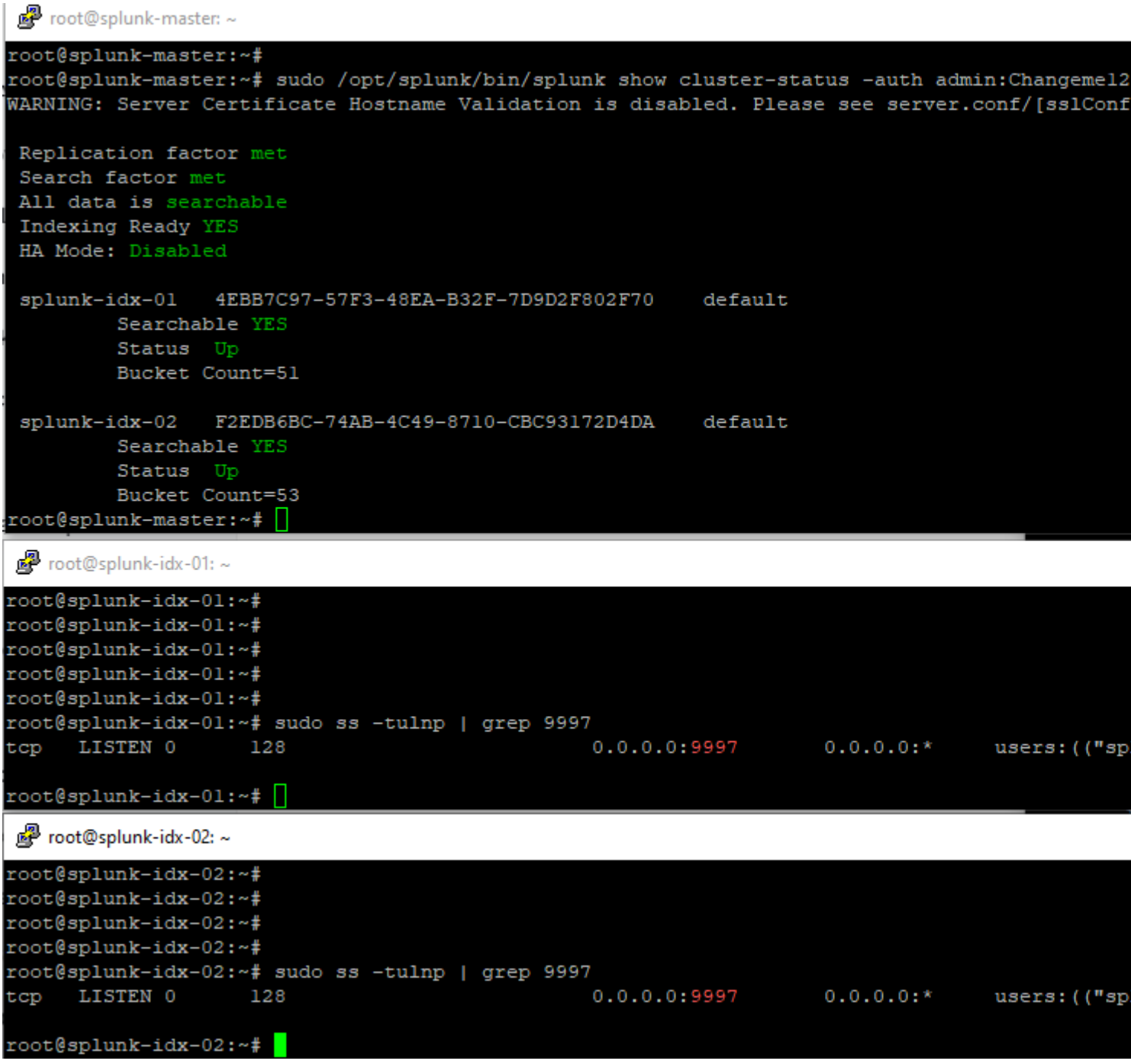
nameserver 127.0.0.53
options edns0
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~# ping -c 2 192.168.1.10  # Ping the Master
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=64 time=1.46 ms

64 bytes from 192.168.1.10: icmp_seq=2 ttl=64 time=2.04 ms

--- 192.168.1.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.460/1.751/2.043/0.291 ms
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~# sudo ss -tulnp | grep 514
udp   UNCONN 0      0                                  0.0.0.0:514        0.0.0.0:*     users:(("rsy
udp   UNCONN 0      0                                     [::]:514           [::]:*     users:(("rsy
tcp   LISTEN 0      25                                 0.0.0.0:514        0.0.0.0:*     users:(("rsy
tcp   LISTEN 0      25                                    [::]:514           [::]:*     users:(("rsy
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~# sudo ls -lh /var/log/remote/
total 15M
-rw-r--r-- 1 syslog adm   26 Oct 19 01:26 new_test.log
-rw-r--r-- 1 syslog adm 195K Oct 20 02:59 splunk-rsyslog01.log
-rw-r--r-- 1 syslog adm   67 Oct 19 00:04 splunk-sh-01.log
-rw-r--r-- 1 syslog adm  15M Oct 20 02:59 ubuntu-target.log
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~# sudo tail -f /var/log/remote/ubuntu-target.log
Oct 20 02:59:48 ubuntu-target sshd[26247]: Disconnecting authenticating user root 192.168.1.1
Oct 20 02:59:48 ubuntu-target sshd[26247]: PAM 5 more authentication failures; logname= uid=0
Oct 20 02:59:48 ubuntu-target sshd[26247]: PAM service(sshd) ignoring max retries; 6 > 3
Oct 20 02:59:48 ubuntu-target sshd[26255]: pam_unix(sshd:auth): authentication failure; logna
```

```
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~#
root@splunk-rsyslog01:~# sudo /opt/splunkforwarder/bin/splunk list monitor
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Your session is invalid.  Please login.
Splunk username: admin
Password:
Monitored Directories:
        $SPLUNK_HOME/var/log/splunk
                /opt/splunkforwarder/var/log/splunk/audit.log
                /opt/splunkforwarder/var/log/splunk/audit_v2.log
                /opt/splunkforwarder/var/log/splunk/btool.log
                /opt/splunkforwarder/var/log/splunk/conf.log
                /opt/splunkforwarder/var/log/splunk/first_install.log
                /opt/splunkforwarder/var/log/splunk/health.log
                /opt/splunkforwarder/var/log/splunk/license_usage.log
                /opt/splunkforwarder/var/log/splunk/mergebuckets.log
                /opt/splunkforwarder/var/log/splunk/mongod.log
                /opt/splunkforwarder/var/log/splunk/mongod_upgrade.log
                /opt/splunkforwarder/var/log/splunk/remote_searches.log
                /opt/splunkforwarder/var/log/splunk/scheduler.log
                /opt/splunkforwarder/var/log/splunk/search_messages.log
                /opt/splunkforwarder/var/log/splunk/searchhistory.log
                /opt/splunkforwarder/var/log/splunk/splunkd-utility.log
                /opt/splunkforwarder/var/log/splunk/splunkd_access.log
                /opt/splunkforwarder/var/log/splunk/splunkd_ui_access.log
                /opt/splunkforwarder/var/log/splunk/wlm_monitor.log
        $SPLUNK_HOME/var/log/splunk/configuration_change.log
                /opt/splunkforwarder/var/log/splunk/configuration_change.log
        $SPLUNK_HOME/var/log/splunk/license_usage_summary.log
                /opt/splunkforwarder/var/log/splunk/license_usage_summary.log
        $SPLUNK_HOME/var/log/splunk/metrics.log
                /opt/splunkforwarder/var/log/splunk/metrics.log
        $SPLUNK_HOME/var/log/splunk/splunk_instrumentation_cloud.log*
                /opt/splunkforwarder/var/log/splunk/splunk_instrumentation_cloud.log
        $SPLUNK_HOME/var/log/splunk/splunkd.log
                /opt/splunkforwarder/var/log/splunk/splunkd.log
        $SPLUNK_HOME/var/log/watchdog/watchdog.log*
                /opt/splunkforwarder/var/log/watchdog/watchdog.log
        $SPLUNK_HOME/var/run/splunk/search_telemetry/*search_telemetry.json
        $SPLUNK_HOME/var/spool/splunk/tracker.log*
        /var/log/remote/
                /var/log/remote/new_test.log
                /var/log/remote/splunk-rsyslog01.log
                /var/log/remote/splunk-sh-01.log
                /var/log/remote/ubuntu-target.log
Monitored Files:
        $SPLUNK_HOME/etc/splunk.version
root@splunk-rsyslog01:~# sudo /opt/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Active forwards:
        192.168.1.12:9997
Configured but inactive forwards:
        192.168.1.11:9997
root@splunk-rsyslog01:~#
```

Splunk Indexers – Cluster Check



Search Head

```
root@splunk-sh-01: ~
root@splunk-sh-01:~#
root@splunk-sh-01:~#
root@splunk-sh-01:~#
root@splunk-sh-01:~# sudo /opt/splunk/bin/splunk list search-server -auth admin:Changeme123
WARNING: Server Certificate Hostname Validation is disabled. Please see server.conf/[sslConf
erverName for details.
Server at URI "192.168.1.11:8089" with status as "Up"
Server at URI "192.168.1.12:8089" with status as "Up"
root@splunk-sh-01:~#
```

Ubuntu Target

Download UF
wget -O splunkforwarder.deb
'https://download.splunk.com/products/universalforwarder/releases/9.2.1/linux/splunkforwarder-9.2.1-<hash>-linux-2.6-amd64.deb'

Install
sudo dpkg -i splunkforwarder.deb

Start with license acceptance
sudo /opt/splunkforwarder/bin/splunk start --accept-license

Enable boot start
sudo /opt/splunkforwarder/bin/splunk enable boot-start

```
root@ubuntu-target: ~
  GNU nano 4.8                          /opt/splunkforwarder/etc/system/local/outputs.conf
[tcpout]
defaultGroup = relay_server

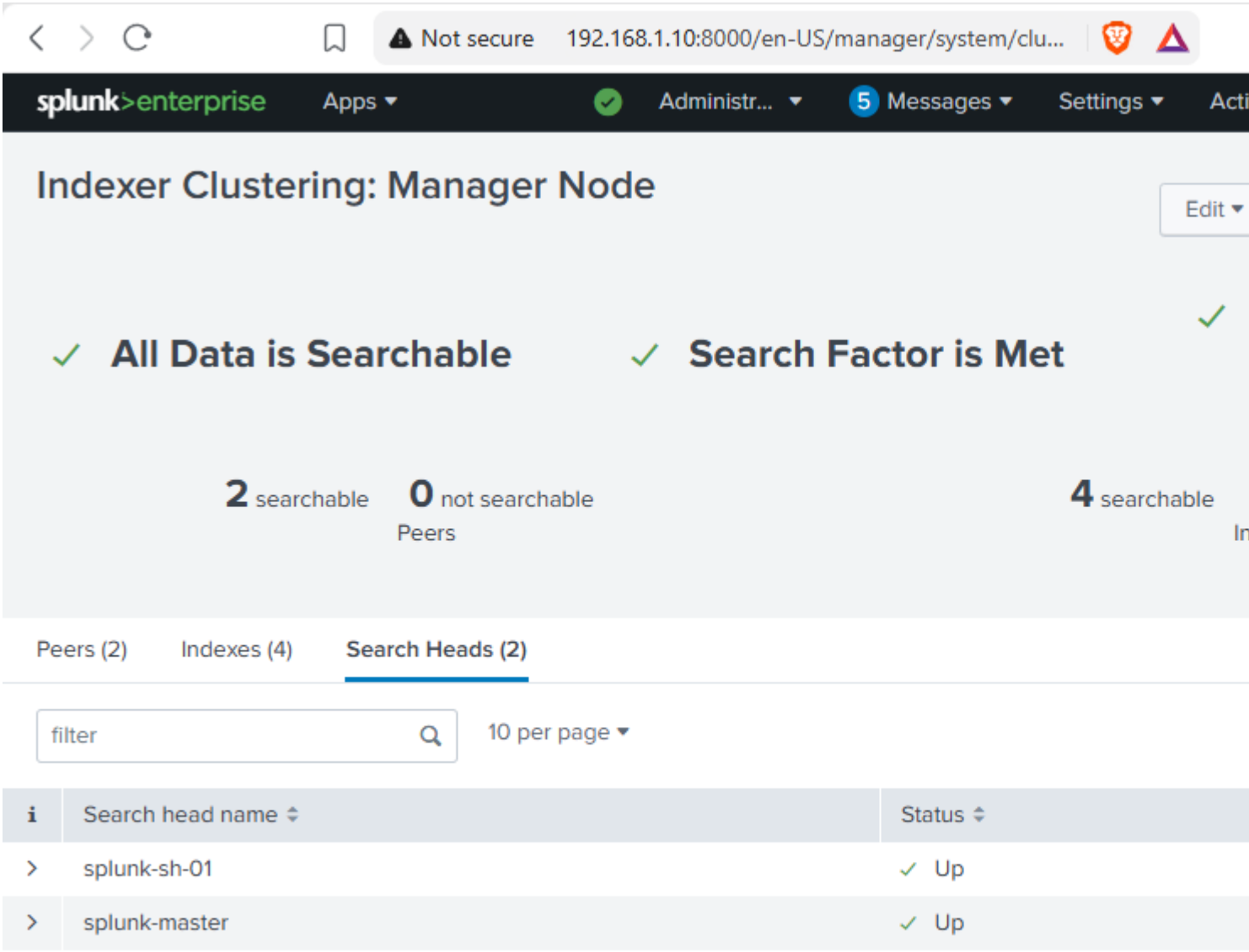
[tcpout:relay_server]
server = 192.168.1.35:9997
```

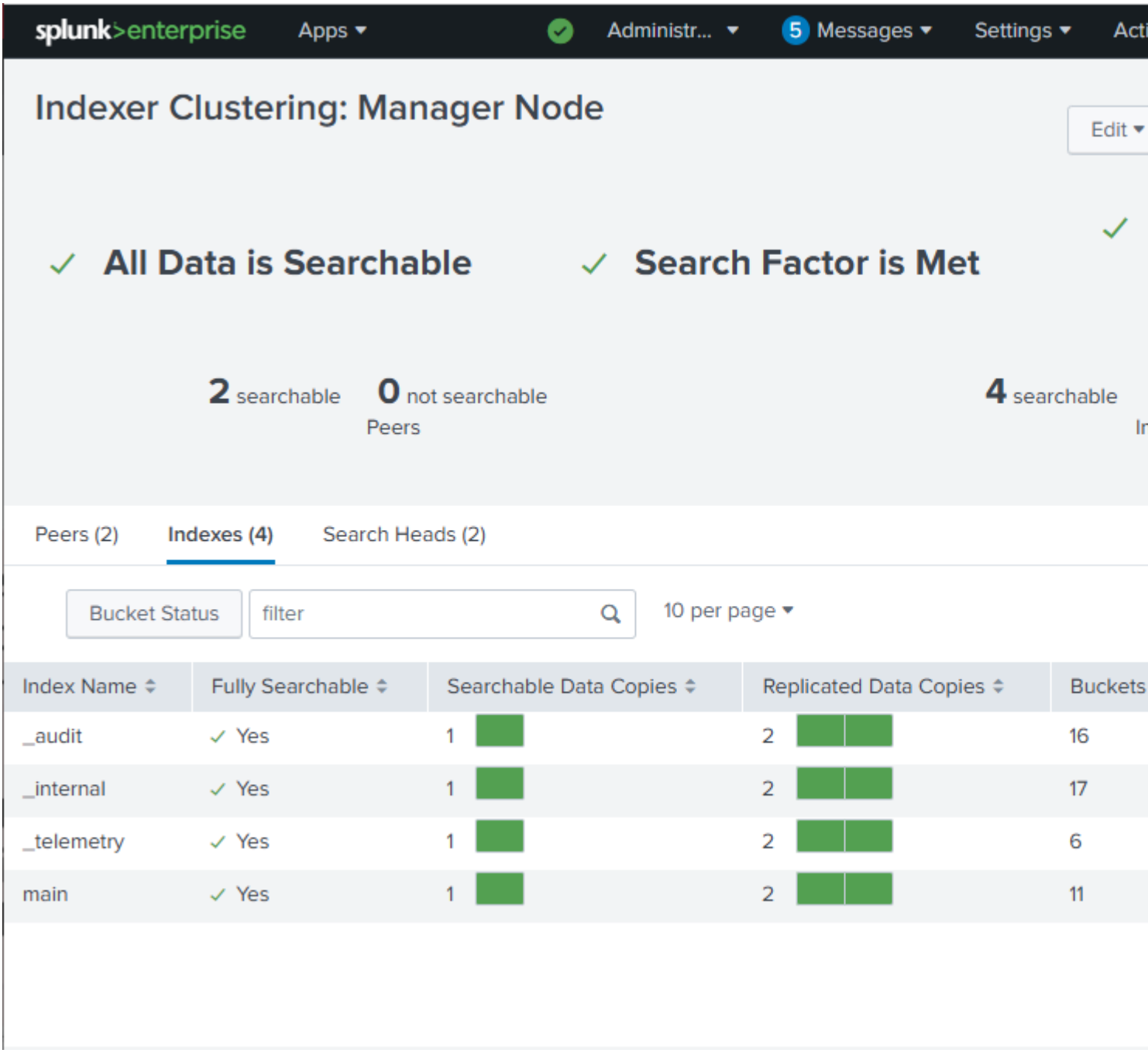
Inputs.conf – Tell the UF what to collect

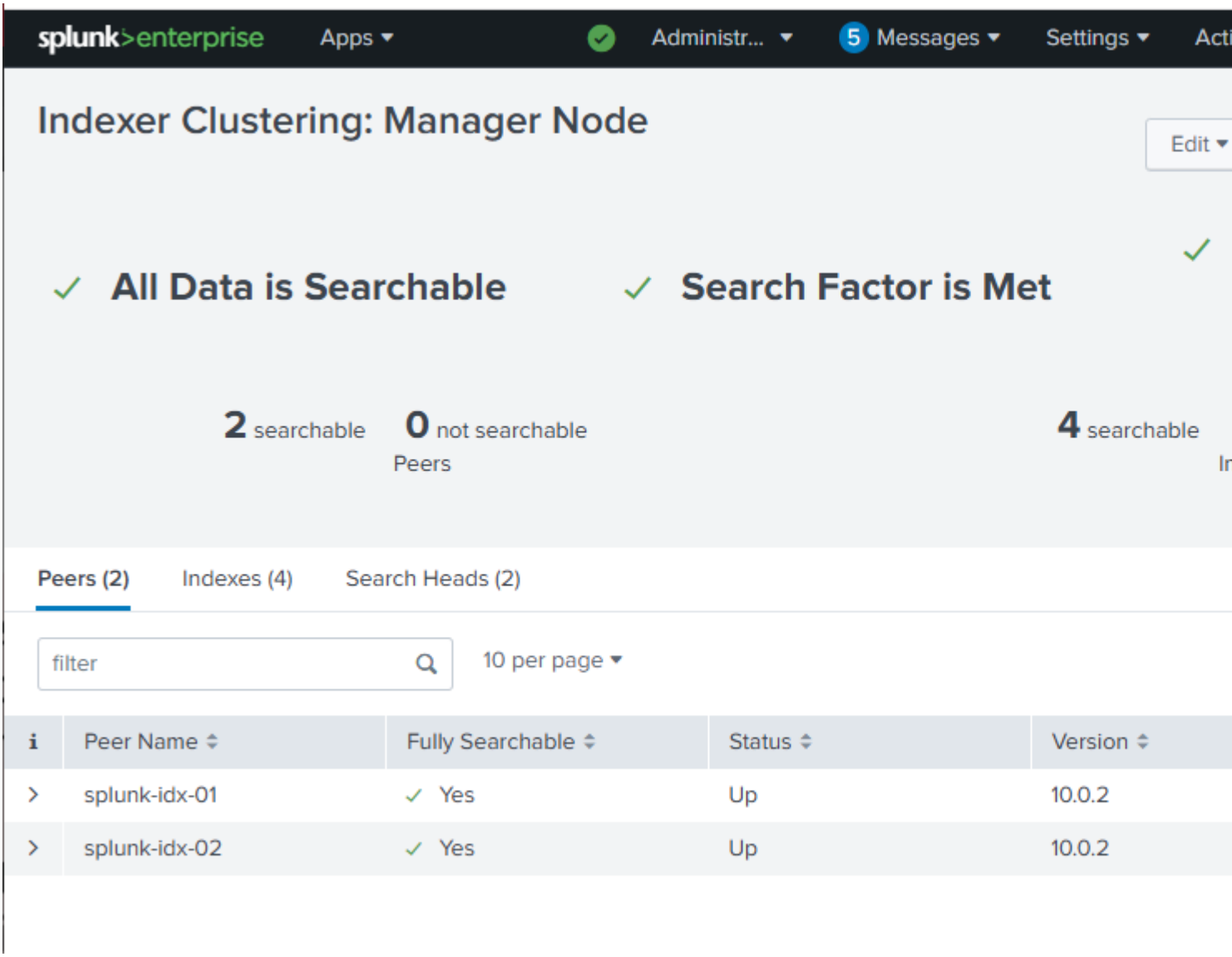
sudo nano /opt/splunkforwarder/etc/system/local/inputs.conf

```
root@ubuntu-target: ~
  GNU nano 4.8                          /opt/splunkforwarder/etc/system/local/inputs.conf
[monitor:///var/log/auth.log]
disabled = false
sourcetype = linux_secure
index = main
```

Splunk Master GUI




splunk>enterprise
Apps
Administr...
5 Messages
Settings
Indexer Clustering: Manager Node
Edit
All Data is Searchable
Search Factor is Met
2 searchable
0 not searchable
Peers
4 searchable
Peers (2)
Indexes (4)
Search Heads (2)
Bucket Status
filter
10 per page
Index Name
Fully Searchable
Searchable Data Copies
Replicated Data Copies
Buckets
_audit
Yes
1
2
16
_internal
Yes
1
2
17
_telemetry
Yes
1
2
6
main
Yes
1
2
11



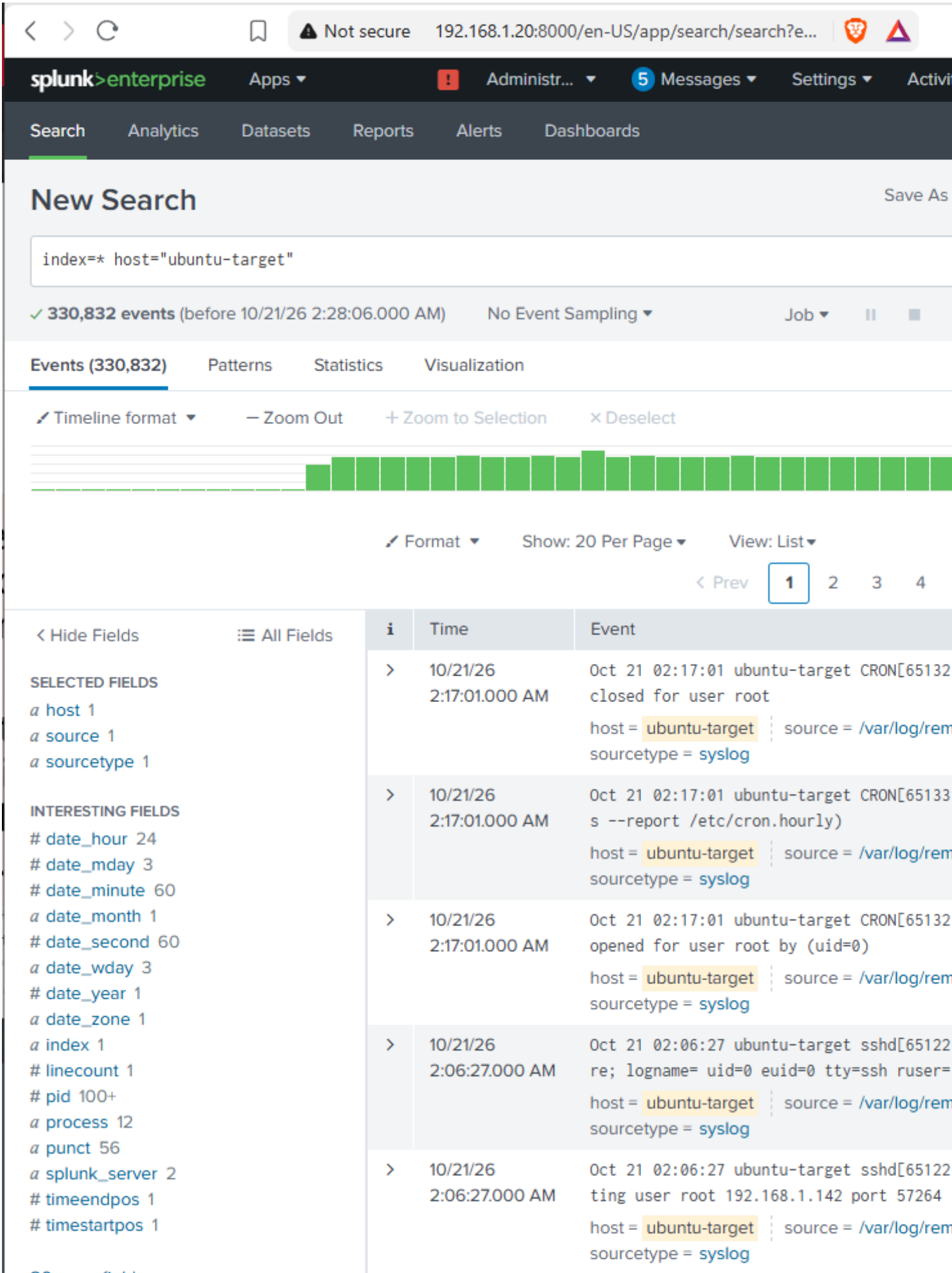
ATTACKER – KALI

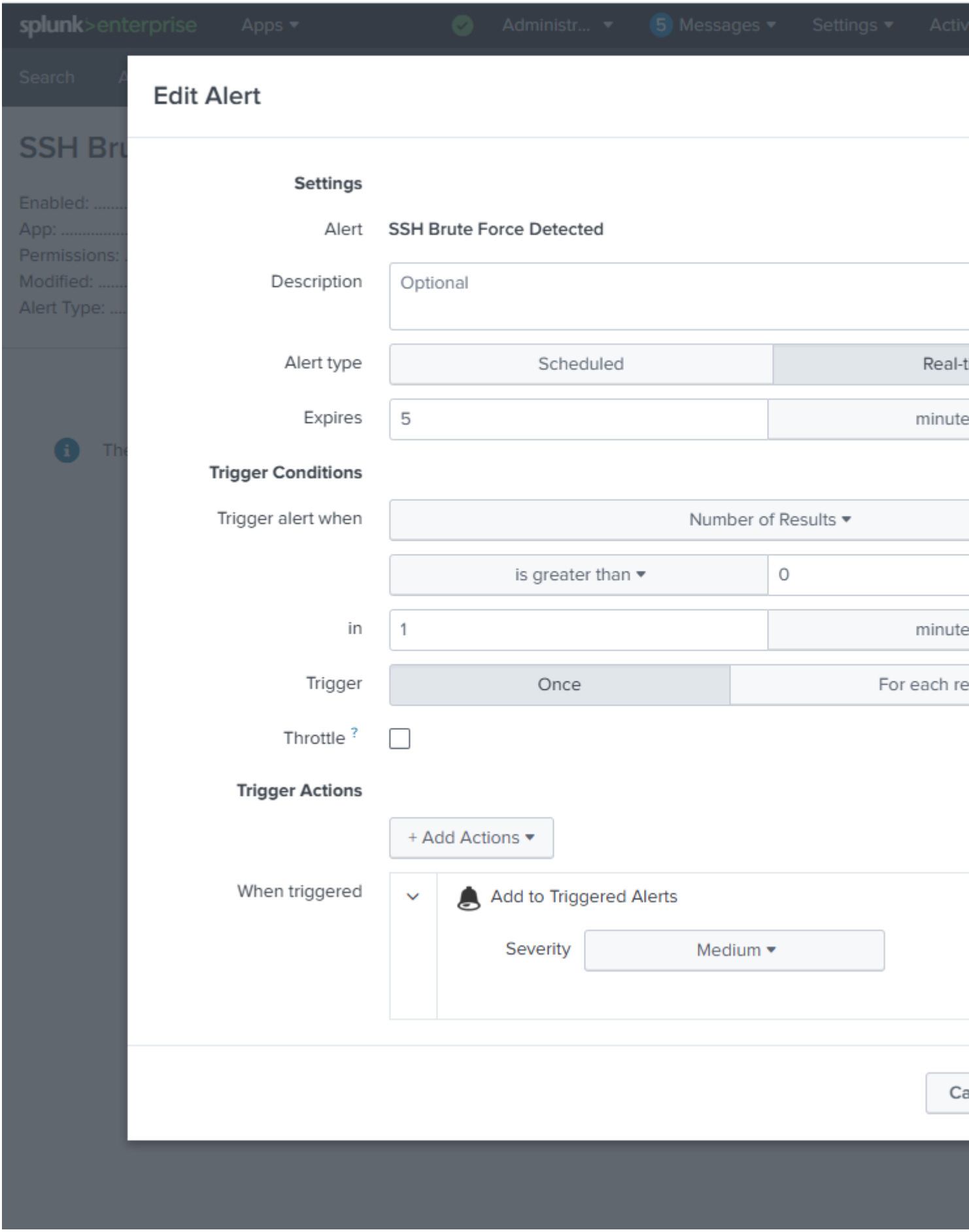
```
┌──(kali㉿kali)-[~]
└─$ hydra -l root -P /usr/share/wordlists/rockyou.txt -t 4 ssh://192.168.1.50
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizati
n-binding, these *** ignore laws and ethics anyway).

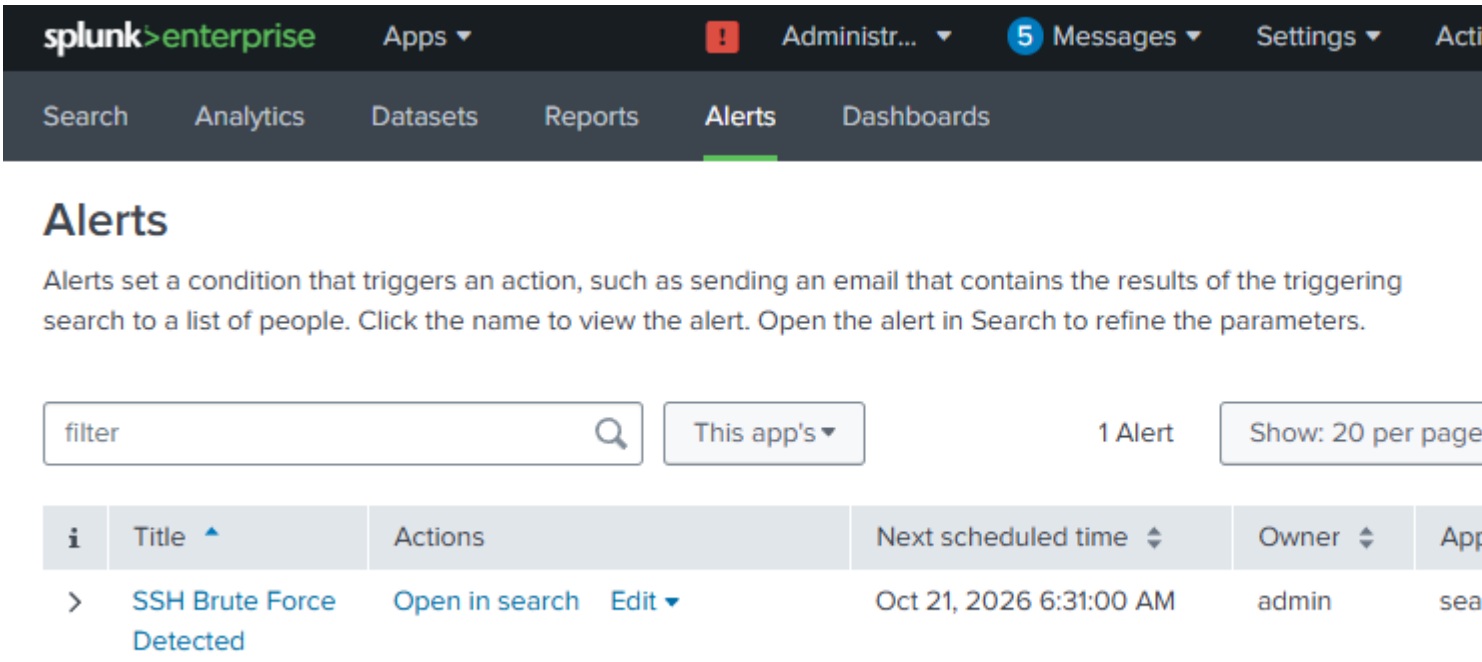
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-10-20 14:01:51
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session fou
ore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://192.168.1.50:22/
[STATUS] 69.00 tries/min, 69 tries in 00:01h, 14344330 to do in 3464:49h, 4 active
[STATUS] 71.33 tries/min, 214 tries in 00:03h, 14344185 to do in 3351:27h, 4 active
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.

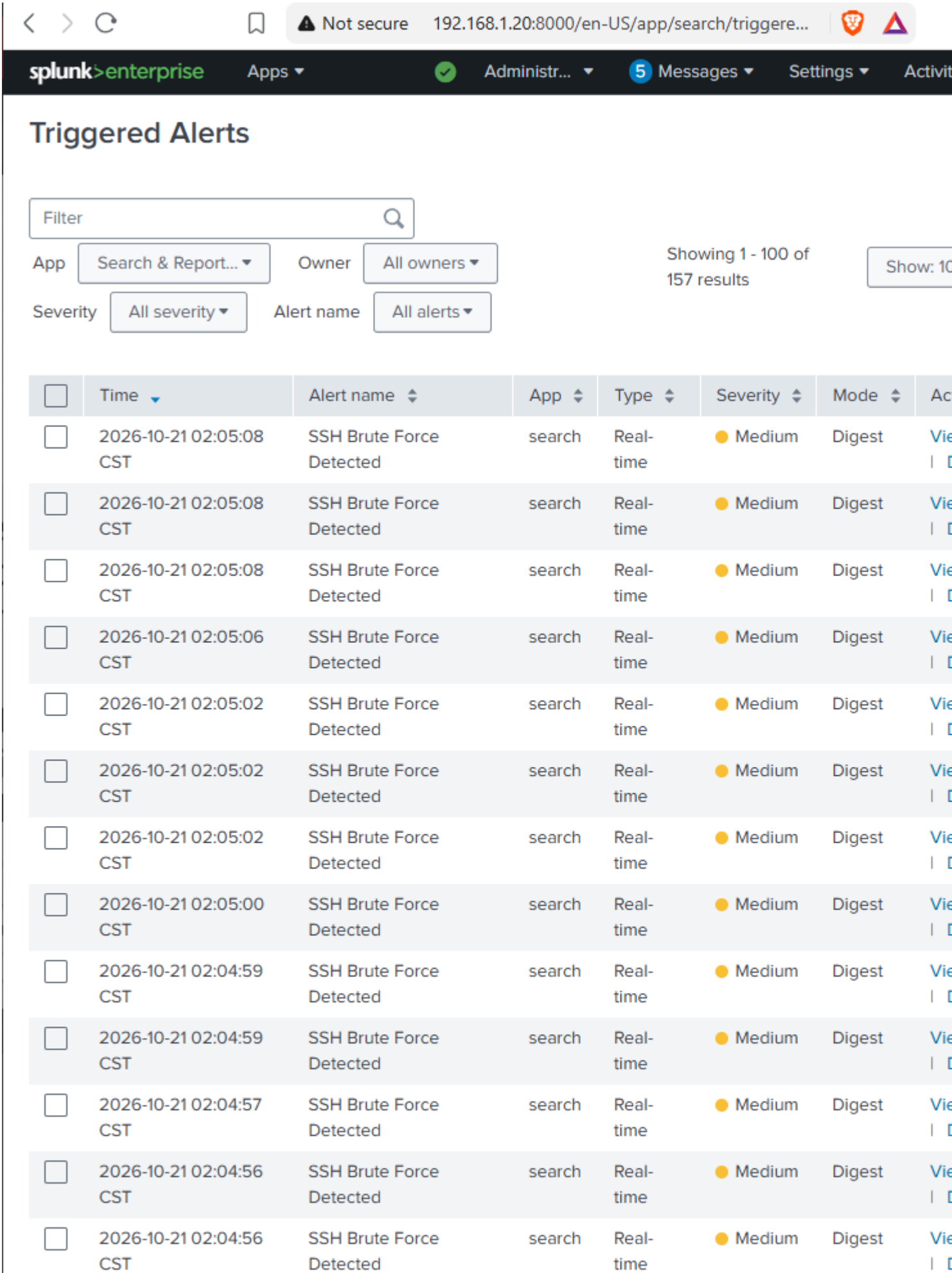
┌──(kali㉿kali)-[~]
└─$
```

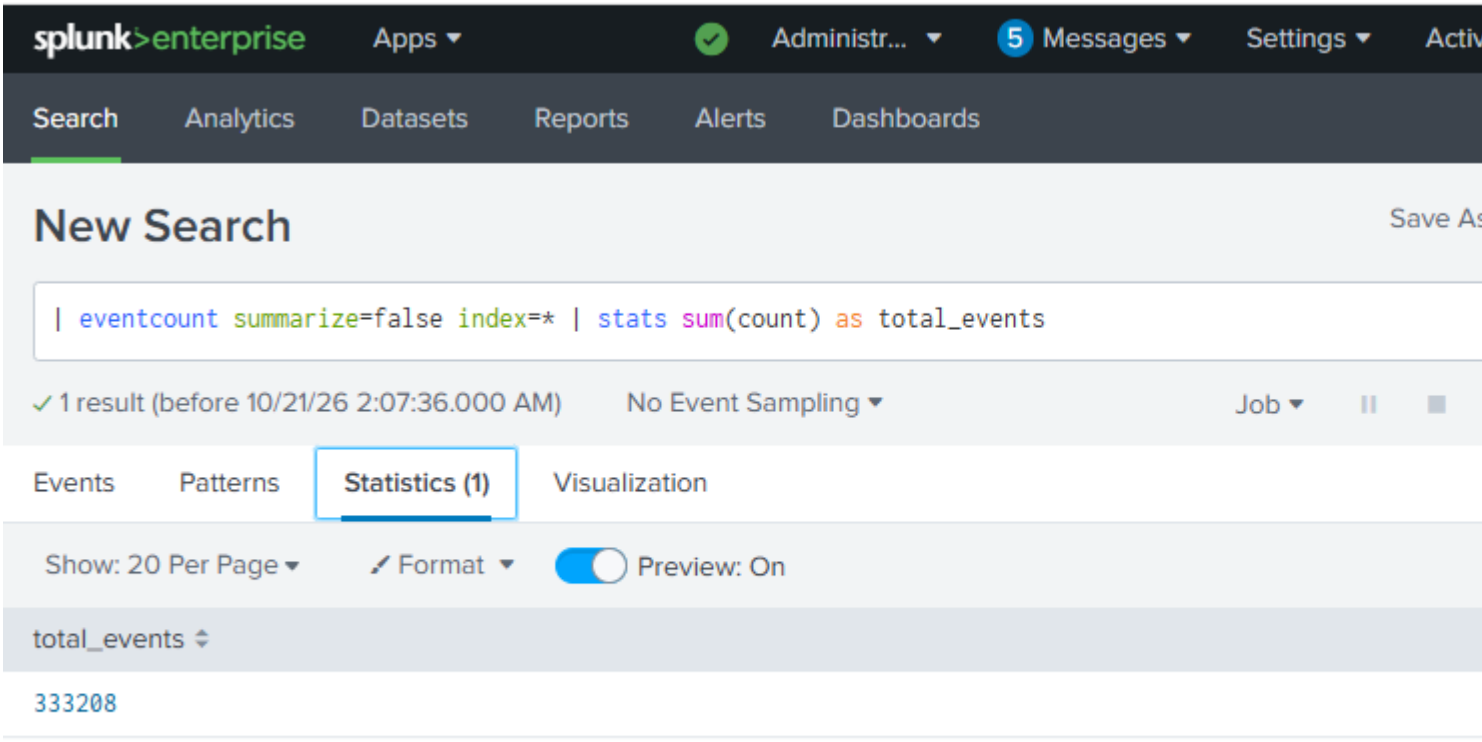
Search Head GUI


Not secure
192.168.1.20:8000/en-US/app/search/search?e...
splunk>enterprise
Apps
Administr...
5 Messages
Settings
Search
Analytics
Datasets
Reports
Alerts
Dashboards
New Search
Save As
index=* host="ubuntu-target"
330,832 events (before 10/21/26 2:28:06.000 AM)
No Event Sampling
Job
Events (330,832)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
Format
Show: 20 Per Page
View: List
Prev
1
2
3
4
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
date_hour 24
date_mday 3
date_minute 60
a date_month 1
date_second 60
a date_wday 3
date_year 1
a date_zone 1
a index 1
linecount 1
pid 100+
a process 12
a punct 56
a splunk_server 2
timeendpos 1
timestartpos 1
i
Time
Event
10/21/26 2:17:01.000 AM
Oct 21 02:17:01 ubuntu-target CRON[65132
closed for user root
host = ubuntu-target
source = /var/log/rem
sourcetype = syslog
10/21/26 2:17:01.000 AM
Oct 21 02:17:01 ubuntu-target CRON[65133
s --report /etc/cron.hourly)
host = ubuntu-target
source = /var/log/rem
sourcetype = syslog
10/21/26 2:17:01.000 AM
Oct 21 02:17:01 ubuntu-target CRON[65132
opened for user root by (uid=0)
host = ubuntu-target
source = /var/log/rem
sourcetype = syslog
10/21/26 2:06:27.000 AM
Oct 21 02:06:27 ubuntu-target sshd[65122
re; logname= uid=0 euid=0 tty=ssh ruser=
host = ubuntu-target
source = /var/log/rem
sourcetype = syslog
10/21/26 2:06:27.000 AM
Oct 21 02:06:27 ubuntu-target sshd[65122
ting user root 192.168.1.142 port 57264
host = ubuntu-target
source = /var/log/rem
sourcetype = syslog


splunk>enterprise
Apps
Administr...
5 Messages
Settings
Search
SSH Bru
Enabled:
App:
Permissions:
Modified:
Alert Type:
Edit Alert
Settings
Alert
SSH Brute Force Detected
Description
Optional
Alert type
Scheduled
Real-t
Expires
5
minute
Trigger Conditions
Trigger alert when
Number of Results
is greater than
0
in
1
minute
Trigger
Once
For each re
Throttle
Trigger Actions
+ Add Actions
When triggered
Add to Triggered Alerts
Severity
Medium
Ca


splunk>enterprise
Apps
Administr...
5 Messages
Settings
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Alerts
Alerts set a condition that triggers an action, such as sending an email that contains the results of the triggering search to a list of people. Click the name to view the alert. Open the alert in Search to refine the parameters.
filter
This app's
1 Alert
Show: 20 per page
Title
Actions
Next scheduled time
Owner
SSH Brute Force Detected
Open in search
Edit
Oct 21, 2026 6:31:00 AM
admin


Not secure
192.168.1.20:8000/en-US/app/search/triggere...
splunk>enterprise
Apps
Administr...
5 Messages
Settings
Activit
Triggered Alerts
Filter
App
Search & Report...
Owner
All owners
Showing 1 - 100 of 157 results
Show: 10
Severity
All severity
Alert name
All alerts
Time
Alert name
App
Type
Severity
Mode
Ac
2026-10-21 02:05:08 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:05:08 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:05:08 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:05:06 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:05:02 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:05:02 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
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2026-10-21 02:05:02 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:05:00 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:04:59 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:04:59 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:04:57 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:04:56 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie
2026-10-21 02:04:56 CST
SSH Brute Force Detected
search
Real-time
Medium
Digest
Vie



Here is the clean, working configuration for all nodes. Do this on each VM:

1. On Splunk-Master (192.168.1.10)
bash
sudo nano /opt/splunk/etc/system/local/server.conf
Make sure the bottom looks like this:
ini
[clustering]
mode = manager
pass4SymmKey = SuperSecretClusterKey123
Save, exit, and run sudo chown splunk:splunk /opt/splunk/etc/system/local/server.conf

2. On Splunk-Idx-01 and Splunk-Idx-02
bash
sudo nano /opt/splunk/etc/system/local/server.conf
Make sure the bottom looks exactly like this:
ini
[replication_port://9887]

[clustering]
mode = peer
manager_uri = https://192.168.1.10:8089
pass4SymmKey = SuperSecretClusterKey123
Save, exit, and run sudo chown splunk:splunk /opt/splunk/etc/system/local/server.conf

3. On Splunk-SH-01 (Search Head)
bash

sudo nano /opt/splunk/etc/system/local/server.conf
Make sure the bottom looks like this:
ini
[clustering]
mode = searchhead
manager_uri = https://192.168.1.10:8089
pass4SymmKey = SuperSecretClusterKey123
Save, exit, and run sudo chown splunk:splunk /opt/splunk/etc/system/local/server.conf

---

3. Restart Everything in Order
To apply the "old" (working) configs, restart the servers in this exact order:
Master: sudo /opt/splunk/bin/splunk restart
Idx-01: sudo /opt/splunk/bin/splunk restart
Idx-02: sudo /opt/splunk/bin/splunk restart
Search Head: sudo /opt/splunk/bin/splunk restart

7. Errors Faced & Troubleshooting Steps

During the deployment, we encountered several critical errors. Here is how we resolved them.

Error 1: Netplan YAML "Tabs not allowed" Error

- **Symptom:** `sudo netplan apply` failed with `Invalid YAML: tabs are not allowed for indent`.
- **Cause:** The `nano` editor inserted tabs instead of spaces.
- **Fix:** Deleted the file contents and manually typed the spaces. YAML strictly requires spaces (usually 2 per indent level). Also, we corrected the interface name from `ens3` to `eth0` based on `ip a` output.

Error 2: Cloud-Init Reverted Hostname

- **Symptom:** After rebooting, the hostname reverted to `kvm` instead of `splunk-idx-01`.
- **Cause:** Cloud-Init overrides manual hostname changes on boot.
- **Fix:** Changed `preserve_hostname: false` to `preserve_hostname: true` in `/etc/cloud/cloud.cfg`.

Error 3: Splunk Web UI Not Loading (Port 8000)

- **Symptom:** Ping worked, but the browser could not reach `http://192.168.1.20:8000`.
- **Cause:** The Splunk service crashed on the Search Head.
- **Fix:** Checked `splunk status`, restarted the service (`splunk restart`), and verified with `ss -tlnp | grep 8000`. Also checked disk space (`df -h`).

Error 4: Forwarder Download 404 Error

- **Symptom:** `wget` failed with `HTTP request sent, awaiting response... 404 Not Found`.
- **Cause:** The URL contained a literal `<hash>` placeholder instead of the actual build hash.
- **Fix:** Used the `latest` URL or the exact build hash for version 9.4.0: `splunkforwarder-9.4.0-6b4ebe426ca6-linux-amd64.deb`.

Error 5: Data Stuck / 0 Events in Splunk (The Cascade Failure)

- **Symptom:** Splunk search showed 0 events. The Forwarder showed "Active forwards" but data never reached the Indexer disk.
- **Cause & Fix (Multiple Issues):**

17. **Missing `outputs.conf`:** The forwarder on `ubuntu-target` was missing its `outputs.conf` file, so it didn't know where to send data. (Created the file pointing to `192.168.1.35:9997`).
18. **Rsyslog not forwarding `auth.log`:** Ubuntu was logging SSH attacks locally but not sending them over the network. (Added `*.* @@192.168.1.35:514` to `/etc/rsyslog.d/50-default.conf`).
19. **Wrong Attack IP:** Hydra was attacking `192.168.1.19` instead of `192.168.1.50`. (Corrected the Hydra command).
20. **File Permissions:** The Forwarder (running as `splunkfwd`) couldn't read Rsyslog files (owned by `syslog`). (Fixed with `chmod 755` and `chmod 644`).
21. **Fishbucket Confusion:** The forwarder remembered old files and refused to re-read them. (Cleared `/opt/splunkforwarder/var/lib/splunk/fishbucket/*` and restarted).
22. **Missing Cluster Peer:** The Cluster Manager was blocking data because `splunk-idx-02` had not joined the cluster. (Ran `edit cluster-config` on Idx-02 and restarted).

Error 6: "No such file or directory" on `grep` Search

- **Symptom:** `sudo grep -r "TEST" /opt/splunk/var/lib/splunk/` returned nothing on the Indexer, despite Splunk Web showing data.
- **Cause:** Splunk compresses raw data into `.tsidx` and `.zst` files. Standard Linux `grep` cannot read inside compressed Splunk buckets.
- **Fix:** Used Splunk's native CLI search instead: `splunk search 'index=* "TEST"'` or used the Web UI.

Error 7: Data Written to Wrong Log File on Relay

- **Symptom:** Hydra attacks were appearing in `splunk-rsyslog01.log` instead of `ubuntu-target.log`.
- **Cause:** Running the `logger -n 127.0.0.1` command on the target caused Rsyslog to tag the source IP as `127.0.0.1` (localhost), routing it to the relay's own filename.
- **Fix:** Avoided using `127.0.0.1` for network tests. Used the actual target IP (`192.168.1.50`) to ensure Rsyslog tagged the hostname correctly.

End of Document